

SIMATS ENGINEERING
ASSESSMENT TOOL 3: Application Based Questions

COURSE NAME: Cloud Computing and Big Data Analytics **COURSE CODE:** CSA15

COURSE OUTCOME COVERED

CO2: Analyze virtualization architectures to identify performance and security issues in cloud computing environments. (BL3)

Dave's Taxonomy: Manipulation (Level 2)
Question Count: 5

Total Marks: 25

Code of Conduct

I **Dilli Babu/192521491** (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: **Dilli Babu**

Assessment Tasks

Application Based Questions

1. A company experiences unpredictable traffic spikes during seasonal sales. Explain how cloud auto-scaling and load balancing can handle this demand efficiently. How does this approach reduce downtime and improve user experience? What role do monitoring tools play in this setup? Relate your answer to a real-world e-commerce scenario.
2. An organization wants to store large volumes of backup data securely in the cloud. Discuss how cloud storage solutions ensure durability and data redundancy. How can encryption and access control protect sensitive backup data? Explain the cost advantages compared to traditional storage systems. Provide an example of a disaster recovery use case.
3. A mobile app company plans to deploy its application globally. Explain how content delivery networks (CDNs) improve performance and reduce latency. How do edge servers enhance user experience across regions? Discuss how cloud providers support global availability. Relate this to a real-time streaming or gaming application.

4. A financial institution requires strict compliance and data security in cloud usage.
Explain how cloud providers support regulatory compliance and auditing. What role do logging, monitoring, and encryption play in security? How can the company ensure data sovereignty requirements are met? Give an example involving sensitive financial transactions.
5. A development team wants to adopt containerization for application deployment.
Explain how containers differ from virtual machines in terms of performance and resource usage. How does orchestration (like Kubernetes) simplify deployment and scaling? Discuss portability benefits across different cloud environments.
Provide an example of a microservices-based application.

Application Based Questions Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Relevance to Application	25%	Response directly addresses the given use case with strong relevance	Mostly relevant response	Partially relevant	Weak relevance
Use of Cloud Concepts	25%	Applies appropriate concepts accurately	Mostly accurate application	Basic application only	Incorrect concept usage
Practical Insight	20%	Shows strong real-world understanding	Shows reasonable practical understanding	Limited practical insight	No practical insight
Completeness	15%	Response covers all required aspects	Covers most aspects	Covers only basic aspects	Incomplete response
Clarity	15%	Clear and concise answer	Generally clear	Some confusion in expression	Poor clarity

1. Cloud Auto-Scaling and Load Balancing for Seasonal Traffic

Introduction

During seasonal sales such as **Black Friday, Diwali, or Amazon Prime Day**, website traffic can increase dramatically. Cloud computing uses **Auto-Scaling** and **Load Balancing** to handle these sudden traffic spikes efficiently.

Cloud Auto-Scaling

- Automatically increases or decreases computing resources based on traffic.
- Adds new virtual servers during high demand.
- Removes unnecessary servers when traffic decreases.
- Ensures optimal performance while minimizing costs.

Load Balancing

- Distributes incoming user requests evenly across multiple servers.
- Prevents any single server from becoming overloaded.
- Redirects traffic to healthy servers if one fails.
- Maintains continuous application availability.

Reducing Downtime and Improving User Experience

- Prevents website crashes during peak traffic.
- Reduces page loading time.
- Ensures uninterrupted shopping experience.
- Improves customer satisfaction and sales.
- Maintains high application availability.

Role of Monitoring Tools

Monitoring tools continuously observe:

- CPU utilization
- Memory usage
- Network traffic
- Response time
- Server health

- Error rates

Examples:

- Amazon CloudWatch
- Azure Monitor
- Google Cloud Monitoring

These tools automatically trigger scaling actions when predefined thresholds are exceeded.

Real-World Example

Amazon experiences millions of users during Prime Day. Auto-scaling launches additional servers automatically, while load balancers distribute customer requests evenly. Monitoring tools detect traffic increases and initiate scaling, ensuring the website remains responsive without downtime.

2. Cloud Storage for Secure Backup

Introduction

Organizations store large amounts of backup data in cloud storage because it provides high durability, security, and scalability.

Durability and Data Redundancy

Cloud providers protect data by:

- Storing multiple copies of data.
- Replicating data across different servers.
- Using multiple data centers.
- Automatically recovering lost data if hardware fails.

This ensures very high data durability (often **99.999999999%**).

Encryption

Cloud storage protects data through:

- Encryption during transmission (TLS/SSL)
- Encryption while stored (AES-256)
- Secure key management

Even if unauthorized users access stored files, encrypted data remains unreadable.

Access Control

Security is enhanced using:

- Identity and Access Management (IAM)
- Multi-Factor Authentication (MFA)
- Role-Based Access Control (RBAC)
- Access policies and permissions

Only authorized users can access backup data.

Cost Advantages

Compared to traditional storage:

- No hardware purchase.
- No maintenance costs.
- Pay only for storage used.
- Easy scalability.
- Reduced electricity and cooling expenses.

Disaster Recovery Example

A company stores backups in cloud storage. When a flood destroys its local data center, backup data stored in the cloud is restored within hours, allowing business operations to continue with minimal downtime.

3. Global Application Deployment Using CDN

Introduction

A Content Delivery Network (CDN) improves application performance by delivering content from servers located close to users.

How CDN Reduces Latency

CDNs:

- Cache static content.
- Reduce network distance.
- Deliver data from nearby servers.

- Decrease page loading time.
- Lower bandwidth consumption.

Role of Edge Servers

Edge servers are geographically distributed.

Benefits:

- Faster content delivery.
- Reduced response time.
- Lower server workload.
- Better performance for international users.
- Improved application availability.

Cloud Provider Global Availability

Major cloud providers operate:

- Multiple regions
- Availability Zones
- Global networks
- Automatic failover systems

This enables users worldwide to access applications quickly.

Real-Time Streaming Example

Netflix uses CDNs worldwide. Videos are cached on edge servers near users, reducing buffering and improving streaming quality regardless of user location.

4. Cloud Security and Compliance in Financial Institutions

Introduction

Financial institutions must protect sensitive customer data while complying with government regulations.

Regulatory Compliance

Cloud providers support standards such as:

- ISO 27001
- PCI-DSS

- SOC 2
- GDPR
- HIPAA (for healthcare)

These certifications ensure secure cloud operations.

Role of Logging

Logging records:

- User activities
- Login attempts
- File access
- Configuration changes
- Security events

Logs help during audits and forensic investigations.

Role of Monitoring

Monitoring continuously checks:

- Suspicious activities
- Network attacks
- Resource usage
- Unauthorized access attempts

Alerts are generated immediately when threats are detected.

Role of Encryption

Encryption protects:

- Data in transit
- Data at rest
- Database information
- Financial records

Only authorized users with proper keys can access sensitive information.

Data Sovereignty

Organizations ensure compliance by:

- Choosing cloud regions within required countries.
- Restricting cross-border data movement.
- Following national data protection laws.

Financial Transaction Example

An online banking system encrypts every transaction, stores audit logs for compliance, continuously monitors suspicious activities, and keeps customer financial records within national data centers to satisfy regulatory requirements.

5. Containerization and Kubernetes

Introduction

Containerization packages applications with all required dependencies, allowing consistent execution across different environments.

Containers vs Virtual Machines

Feature	Containers	Virtual Machines
Startup Time	Seconds	Minutes
Resource Usage	Low	High
Operating System	Shares Host OS	Separate Guest OS
Performance	Faster	Slower
Portability	Very High	Moderate

Benefits of Containers

- Lightweight
- Faster deployment
- Better resource utilization
- Easy scaling
- Consistent execution across environments

Role of Kubernetes

Kubernetes automates:

- Container deployment
- Load balancing
- Auto-scaling
- Service discovery
- Self-healing
- Rolling updates
- High availability

This reduces manual management and improves reliability.

Portability Across Cloud Providers

Containers can run consistently on:

- Amazon Web Services (AWS)
- Microsoft Azure
- Google Cloud Platform (GCP)
- Private cloud
- Hybrid cloud
- On-premises infrastructure

This avoids vendor lock-in and simplifies migration.

Microservices Example

An **e-commerce application** can be divided into multiple containerized services:

- User Authentication Service
- Product Catalog Service
- Shopping Cart Service
- Payment Service
- Inventory Service
- Order Management Service
- Notification Service

Each microservice runs independently in its own container and can be scaled individually using Kubernetes, improving application performance, availability, and maintainability.